

Strictly nef divisors on singular varieties, I

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Today's talk is based on the joint work with Jie Liu, Wenhao Ou, Juanyong Wang and Xiaokui Yang.

Background & Motivation

We start from a projective variety $X/\mathbb{C}$.

Definition (Strictly nef)

A $\mathbb{Q}$ -Cartier divisor L on X is said to be **strictly nef**, if $L \cdot C > 0$ for every (integral irreducible) curve C on X .

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- If X is a curve, then every strictly nef divisor L on X is ample.
- If (X, Δ) is an lc pair and the log canonical divisor $K_X + \Delta$ is strictly nef, then the abundance conjecture and the projection formula would imply $K_X + \Delta$ is ample. In particular, if $\dim X \leq 3$, then $K_X + \Delta$ being strictly nef forces $K_X + \Delta$ to be ample.

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- The pullback of a strictly nef (resp. non-ample) divisor along a finite morphism, is still strictly nef (resp. non-ample).

Background & Motivation

Main question

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Strictly nef non-ample divisors do exist.

Example B (Mumford's example)

Let C be a smooth projective curve of genus ≥ 2 . Then there exists a vector bundle E of rank two over C such that the tautological bundle $\mathcal{O}_{\mathbb{P}(E)}(1)$ on the projectivization $\mathbb{P}(E)$ is strictly nef, but not ample.

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The following conjecture is the initial point of this project.

Campana-Peternell's conjecture (1991)

Let X be a smooth projective variety. If $-K_X$ is strictly nef, then X is Fano, i.e., $-K_X$ is ample.

Background & Motivation

Serrano proposed the generalized conjecture, predicting the ampleness of a strictly nef divisor after a suitable deformation in the direction of K_X .

Serrano's conjecture (1995)

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Remarks on the two conjectures

- Serrano's conjecture for the case $\dim X = n \Rightarrow$ CP's conjecture for the case $\dim X = n$.

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- Serrano's conjecture for the case $\dim X \leq n$ is important to the proof of CP's conjecture for the case $\dim X = n + 1$ as detected by Serrano.

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- Serrano's conjecture for the case $\dim X = n \Rightarrow$ CP's conjecture for the case $\dim X = n$.
- Serrano's conjecture for the case $\dim X \leq n$ is important to the proof of CP's conjecture for the case $\dim X = n + 1$ as detected by Serrano.
- As observed by Liu, a positive answer to the log version of Serrano's conjecture in $\dim X = 3$ will give a positive answer to CP's conjecture in $\dim X = 4$. [the singular \(pair\) setting?](#)

Background & Motivation

Question (Singular version of Campana-Peternell's conjecture)

Let (X, Δ) be a projective klt pair with the anti-log canonical divisor $-(K_X + \Delta)$ being strictly nef. Is $-(K_X + \Delta)$ ample?

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Let (X, Δ) be a projective klt pair, and L_X a strictly nef $\mathbb{Q}$ -divisor on X . Is $K_X + \Delta + tL_X$ ample for $t \gg 1$?

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It has a positive answer on surfaces by studying the generalized pairs.

Theorem (surface case; Han-Liu, 2020)

Let (X, Δ) be a projective lc surface pair, with L_X a strictly nef Cartier divisor on X . Then $K_X + \Delta + tL_X$ is ample for every real number $t > 3$.

Our main focus today is on the first question. We shall study Serrano's conjecture next time.

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Let X be a canonical 3-fold with strictly nef $-K_X$. Then $-K_X$ is ample.

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Theorem (Liu, 2021)

A smooth projective 4-fold X with $c_1^2(X) \cdot c_2(X) \neq 0$ and $-K_X$ being strictly nef is a Fano fourfold.

Structure theorem of (X, Δ) with nef $-(K_X + \Delta)$.

First, we list some properties and historical remarks here.

- By (BDPP, 2013), if $-(K_X + \Delta)$ is strictly nef, then X is uniruled.
- For a projective variety X , there is a dominant rational map $X \dashrightarrow Y$ onto a projective variety Y (unique up to birational equivalence) such that the general fibre is rationally connected and Y is not uniruled. We say that such a map is the MRC fibration of X .
- Based on Mori's bend and break, (Kollár-Miyaoka-Mori, 1992) and (Campana, 1992) showed that Fano manifolds are rationally connected.
- (Zhang, 2006) and (Hacon-McKernan, 2007) generalized the rational connectedness of Fano manifolds to varieties of weak Fano type.
- Beauville-Bogomolov decomposition: Calabi-Yau manifolds ($K_X \equiv 0$) are decomposed, up to a finite étale cover, into the product of abelian varieties, strict Calabi-Yau manifolds, and hyperkähler manifolds.

Structure theorem of (X, Δ) with nef $-(K_X + \Delta)$.

- (Greb-Kebekus-Peternell, 2016), (Druel, 2018), (Höring-Peternell, 2019), (Greb-Guenancia-Kebekus, 2019), (Campana-Păun 2019), (Campana, 2020) generalized BB-decomp. to singular klt Calabi-Yau varieties.
- (Bakker-Guenancia-Lehn, 2021) dealt with the decomposition for (non-projective) K -trivial compact Kähler varieties.
- There are some progress in positive characteristics, but more technical conditions were imposed; see (Ejiri, 2019), (Ejiri-Gongyo, 2019), (Patakfalvi-Zdanowicz, 2020).
- Back to characteristic zero, (Cao, 2016) and (Cao-Höring, 2017) showed that the smooth projective varieties with nef anti-canonical divisor are constructed by rationally connected manifolds and Calabi-Yau manifolds.
- (Wang, 2020) and (Matsumura-Wang, 2021) constructed the structure theorem of a projective klt pair (X, Δ) with nef $-(K_X + \Delta)$.

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Theorem (Matsumura-Wang, 2021)

Let (X, Δ) be a projective klt pair with $-(K_X + \Delta)$ being nef. Then, there exists a finite quasi-étale cover $\nu : X' \rightarrow X$ such that:

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- (1) X' admits a holomorphic MRC fibration $\psi : X' \rightarrow Y$.
- (2) Y is a projective klt variety with $K_Y \equiv 0$.

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- (2) Y is a projective klt variety with $K_Y \equiv 0$.
- (3) ψ is locally constant with respect to the pair (X', Δ') , where Δ' is the $\mathbb{Q}$ -Weil divisor defined by the pull-back $\Delta' := \nu^* \Delta$. To be precise,
 - (a) For any small open set $B \subseteq Y$, we have $(\psi^{-1}(B), \Delta') \cong B \times (F, \Delta'_F)$ over $B \subseteq Y$, where F is the fibre of ψ and $\Delta'_F := \Delta'|_F$.
 - (b) Every component of Δ' is horizontal.
 - (c) There is a representation $\rho : \pi_1(Y) \rightarrow \text{Aut}(F)$ such that Δ'_F is stable under the action of $\pi_1(Y)$, so that $X' \times_Y Y^{\text{univ}} \cong Y^{\text{univ}} \times F$ and

$$(X', \Delta') \cong (Y^{\text{univ}} \times F, \text{pr}_2^* \Delta'_F) / \pi_1(Y)$$

Structure theorem of (X, Δ) with nef $-(K_X + \Delta)$.

With (X, Δ) replaced by its quasi-étale cover, we get a holomorphic locally constant MRC fibration $(X, \Delta) \xrightarrow{f} Y$ with Y being klt and $K_Y \equiv 0$.

$$\begin{array}{ccc}
 X \times_Y Y^{\text{univ}} \simeq Y^{\text{univ}} \times F & \xrightarrow{f^{\text{univ}} = \text{pr}_1} & Y^{\text{univ}} \\
 \downarrow p_X & \square & \downarrow p_Y \\
 X & \xrightarrow{f} & Y.
 \end{array}$$

- The morphism f is locally trivial analytic fibre bundle with the fibre F .
- Every component Δ_i of Δ is horizontal (i.e., $f(\Delta_i) = Y$).
- There is a representation $\rho : \pi_1(Y) \rightarrow \text{Aut}(F)$ and a Weil $\mathbb{Q}$ -divisor Δ_F on F which is invariant under the action of $\pi_1(Y)$, so that $(X, \Delta) \cong (Y^{\text{univ}} \times F)/\pi_1(Y)$ via the diagonal action.

Main results

Question (A weak form of Campana-Peternell's conjecture)

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- Weak Fano type $\Rightarrow$ rationally connected (Zhang, 2006).
- When X is smooth and $\Delta = 0$, applying the inspiring results (Cao, 2016) and (Cao-Hörling, 2017), Li-Ou-Yang gave a positive answer to the above question in 2019.
- As a corollary, (Li-Ou-Yang, 2019) gave an analogue of Mori's result: A smooth projective variety with strictly nef tangent bundle is $\mathbb{P}^n$.
- Inspired by the above joint works, our first main result is to give a positive answer to the weak form of Campana-Peternell's conjecture by applying the work (Wang-Mastumura, 2021).

Main results

Theorem A (Liu-Ou-Wang-Yang-Z., 2021)

Let (X, Δ) be a projective klt pair with strictly nef $-(K_X + \Delta)$. Then X is rationally connected. Moreover, the augmented irregularity $q^\circ(X) = 0$.

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A few remarks on Theorem A:

- Theorem A answers the weak form of the generalized (log version) Campana-Peternell conjecture.
- $q^\circ(X)$ is defined as the maximum of $q(\tilde{X})$ where $\tilde{X} \rightarrow X$ runs over the finite quasi-étale covers of X (Nakayama-Zhang, 2010).
- When $\dim X = 3$, it was proved by (Z., 2021).
- One can further generalize this theorem to the case when $-(K_X + \Delta)$ is **almost strictly nef**, i.e., $-(K_X + \Delta) = \pi^*L_Y$ for some birational morphism $\pi : X \rightarrow Y$ and a strictly nef divisor L_Y on Y .

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Our next main result generalizes Theorem A to the relative case, and analyzes such fibration like Example B.

Main results

Theorem B (Liu-Ou-Wang-Yang-Z., 2021)

Let $f : X \rightarrow Y$ be a fibration from a normal projective variety X to a **smooth** projective variety Y . Suppose that (X, Δ) is a klt pair and the relative anti-log canonical divisor $-(K_{X/Y} + \Delta)$ is strictly nef. Then f is a locally constant fibration with rationally connected fibres and Y is a canonically polarized hyperbolic projective manifold.

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- A projective manifold is said to be **canonically polarized hyperbolic** if K_Y is ample and every holomorphic map $\mathbb{C} \rightarrow Y$ is constant.
- Theorem B is a more general form of Theorem A, while Theorem A is one of the key ingredients during the proof of Theorem B.
- We don't know whether the smoothness of Y can be dropped. The smoothness of Y is to make sure that f is locally constant by (Matsumura-Wang, 2021) and (Campana-Cao-Matsumura, 2019).
- However, if we have already assumed the local constancy of f , then we can remove the smoothness of Y .

Main results

Our third result is to answer the log version of Campana-Peternell's conjecture affirmatively in dimension three.

Theorem C (Liu-Ou-Wang-Yang-Z., 2021)

Let (X, Δ) be a projective klt pair with the anti-log canonical divisor $-(K_X + \Delta)$ being strictly nef. If $\dim X = 3$, then $-(K_X + \Delta)$ is ample.

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- (Uehara, 2000) took the first glance on the singular case when X is a threefold with only canonical singularities and $\Delta = 0$.
- The key point to show Theorem C is to prove that $-(K_X + \Delta)$ is numerically equivalent to an effective divisor.

Proposition (Liu-Ou-Wang-Yang-Z., 2021)

Let (X, Δ) be a projective klt threefold pair, and L_X a strictly nef $\mathbb{Q}$ -divisor on X . Suppose that L_X is numerically equivalent to a non-zero effective divisor. Then $K_X + \Delta + tL_X$ is ample for $t \gg 1$.

Key ideas for the proof of Theorem A

Theorem (Section of the MRC fibration; Liu-Ou-Wang-Yang-Z., 2021)

Let (X, Δ) be a projective klt pair with the anti-log canonical divisor $-(K_X + \Delta)$ being nef. Then up to replacing X with a quasi-étale cover, the MRC fibration $f : X \rightarrow Y$ in [MW21] is a locally constant fibration which admits a section $\sigma : Y \rightarrow X$ such that $\sigma^*(K_X + \Delta) \equiv 0$.

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The above theorem implies our Theorem A:

- We take the (locally constant) MRC fibration $f : X \rightarrow Y$ after replacing X by a quasi-étale cover.
- Suppose the contrary that X is not rationally connected. Then $\dim Y > 0$.
- By the above theorem, up to replacing X by a further quasi-étale cover, our f admits a section $\sigma : Y \rightarrow X$ such that $\sigma^*(K_X + \Delta) \equiv 0$.
- Pick a curve C in Y . By the projection formula,

$$0 = -\sigma^*(K_X + \Delta) \cdot C = -(K_X + \Delta) \cdot \sigma_* C > 0,$$

and we get a contradiction.

Key ideas for the proof of Theorem B

We start from a fibration $f : (X, \Delta) \rightarrow Y$ of a projective klt pair (X, Δ) onto a **smooth** projective manifold Y such that $-(K_{X/Y} + \Delta)$ is nef.

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We start from a fibration $f : (X, \Delta) \rightarrow Y$ of a projective klt pair (X, Δ) onto a **smooth** projective manifold Y such that $-(K_{X/Y} + \Delta)$ is nef.

Technical Lemma 1

Let T be the Zariski closure of the union of curves in X with $-(K_{X/Y} + \Delta)$ -degree 0. If T does not dominate Y , then Y is of general type, and there exists a linear representation $\chi : \pi_1(Y) \rightarrow \mathrm{GL}(r, \mathbb{C})$ whose image is not virtually solvable.

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Lemma 2 (Liu-Ou-Yang, 2020)

Let Y be an irreducible projective variety. If there exists a representation $\rho : \pi_1(Y) \rightarrow \mathrm{GL}(r, \mathbb{C})$ such that its image $\mathrm{im}(\rho)$ is not virtually solvable, then every holomorphic map $\mathbb{C} \rightarrow Y$ is degenerate, i.e., the image is not Zariski dense in Y .

Key ideas for the proof of Theorem B

Theorem (*) (Campana-Cao-Matsumura, 19, Matsumura-Wang, 21)

With the assumption as above, let F be a general fibre of f . Then f is a locally constant fibration with respect to the pair (X, Δ) .

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Now we prove our Theorem B.

- By Thm (*), we are left to show Y is hyperbolic and K_Y is ample.
- By Lemma 1, K_Y is big. Therefore, K_Y is ample $\Leftarrow Y$ does not contain any rational curves $\Leftarrow Y$ is hyperbolic.
- Let $h : \mathbb{C} \rightarrow Y$ be holomorphic and $Z := \overline{h(\mathbb{C})}$. Suppose $\dim Z > 0$. Then $h(\mathbb{C})$ is not degenerate in Z . Let $Z' \rightarrow Z$ be the normalization and $f' : X' \rightarrow Z'$ the induced locally constant fibration.
- By Lemma 1, there exists a representation of $\pi_1(Z')$ whose image is not virtually solvable. On the other hand, the holomorphic map h can be lifted to $h' : \mathbb{C} \rightarrow Z'$; thus, $\overline{h'(\mathbb{C})} = Z'$, contradicting Lemma 2.

Key ideas for the proof of Theorem C

Theorem (Han-Liu, 2020)

Let $(X, \Delta + M_X)$ be a 3-dim projective generalized klt pair such that M_X is an $\mathbb{R}_{>0}$ -linear combination of nef Cartier b-divisors. Suppose that $K_X + \Delta + M_X$ is pseudo-effective and $K_X + M_X$ is not pseudo-effective. Then $K_X + \Delta + M_X$ is numerically equivalent to an effective $\mathbb{R}$ -divisor.

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Proposition (Liu-Ou-Yang-Wang-Z., 2021)

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- A klt threefold pair (X, Δ) with $-(K_X + \Delta)$ being strictly nef.
- Take a $\mathbb{Q}$ -factorial terminalization $(Y, \Gamma) \xrightarrow{\tau} (X, \Delta)$ such that (Y, Γ) is a $\mathbb{Q}$ -factorial terminal pair.
- If $\Gamma \neq 0$, we are done by the above two results. If $\Gamma = 0$, then $\Delta = 0$ and we reduce to the case when X has only canonical singularities.

Open Question

Question 1

Let $f : (X, \Delta) \rightarrow Y$ be a locally constant fibration with $-(K_{X/Y} + \Delta)$ being **strictly** nef. If we remove the condition that Y is smooth, will such f still be locally constant?

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If $-(K_{X/Y} + \Delta)$ is only nef, then the question has a negative answer by considering the terminalization $(X, \Delta) \rightarrow Y$ of a projective klt variety Y . In this case, $-(K_{X/Y} + \Delta)$ is relatively trivial but f is not locally constant. In general, we expect that the strict nefness could restrict and control the behaviour of f .

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Question 1

Let $f : (X, \Delta) \rightarrow Y$ be a locally constant fibration with $-(K_{X/Y} + \Delta)$ being **strictly** nef. If we remove the condition that Y is smooth, will such f still be locally constant?

If $-(K_{X/Y} + \Delta)$ is only nef, then the question has a negative answer by considering the terminalization $(X, \Delta) \rightarrow Y$ of a projective klt variety Y . In this case, $-(K_{X/Y} + \Delta)$ is relatively trivial but f is not locally constant. In general, we expect that the strict nefness could restrict and control the behaviour of f .

Question 2

Let (X, Δ) be a strongly F -regular (or sharply F -pure) pair with strictly nef $-(K_X + \Delta)$. Is X rationally chain connected? Is $-(K_X + \Delta)$ ample?

The surface case is not very hard, but from threefolds, one needs to develop new tools to study this question.

Thank you for your attention!